Solution

Matrix Theory, Quiz 4R — February 20, 2026

Answer the following for the matrix A and vectors $\mathbf{v}$, $\mathbf{w}$:

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 4 \\ -4 \\ 4 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

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(1) Find the reduced row echelon form of the augmented matrix $[A \mid \mathbf{v}\,\mathbf{w}]$.

4@1

(2) Find the determinant, rank and nullity of A.

(3) Find a basis for the column space of A.

(4) Find a basis for the null space of A.

(5) True or False:

(a) A is invertible.

(b) $\mathbf{v}$ is in the column space of A.

(c) $\mathbf{w}$ is in the column space of A.

(d) $\mathbf{w}$ is in the null space of A.

RREF

$$\left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & 4 & 4 \\ 0 & -1 & 1 & -1 & -4 & 4 \\ 1 & 0 & -1 & 0 & 4 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & 4 & 4 \\ 0 & -1 & 1 & -1 & -4 & 4 \\ 0 & 0 & 0 & -2 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & 4 & 4 \\ 0 & 0 & 0 & -2 & 0 & 8 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 4 & 4 \\ 0 & 1 & -1 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 4 & 0 \\ 0 & 1 & -1 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 8 \end{array}\right]$$

(2) $\det A = 0$, $\operatorname{Rank} A = 3$, $\operatorname{Null} A = 1$

(3) c_1, c_2, c_4 of A

(4) $\operatorname{Null\,sp} A = x_3 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$ basis $= \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$

(5) (a) F

(b) T

(c) F

(d) F

Version 4 T

RREF: $\left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 0 & 4 \\ 0 & 1 & -1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$

(5) (a) F

(b) F

(c) T

(d) F

Solution

Matrix Theory, Quiz 4S — February 20, 2026

Answer the following for the matrix A and vectors $\mathbf{v}$, $\mathbf{w}$:

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ -1 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 4 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

6

4@1

(1) Find the reduced row echelon form of the augmented matrix $[A \mid \mathbf{v}\,\mathbf{w}]$.
(2) Find the determinant, rank and nullity of A.
(3) Find a basis for the column space of A.
(4) Find a basis for the null space of A.
(5) True or False:
  (a) A is invertible.
  (b) $\mathbf{v}$ is in the column space of A.
  (c) $\mathbf{w}$ is in the column space of A.
  (d) $\mathbf{w}$ is in the null space of A.

(1)
$$\left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & -1 & 1 \\ 0 & -1 & 1 & -1 & -1 & 1 \\ 1 & 0 & -1 & 0 & 2 & 2 \end{array}\right] \to \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & -1 & 1 \\ 0 & -1 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & -2 & -2 & -2 \end{array}\right] \xrightarrow[-2]{R_4} \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 2 & 4 & 4 \\ 0 & 1 & -1 & -1 & -1 & 1 \\ 0 & 0 & 0 & -2 & -2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow[R_3+2R_4]{R_1-2R_4,\ R_2+R_4} \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 2 & 2 \\ 0 & 1 & -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right] \xrightarrow{R_3 \leftrightarrow R_4} \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 2 & 2 \\ 0 & 1 & -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 4 \end{array}\right] \xrightarrow{\frac{1}{4}R_4} \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

RREF

(2) $\det A = 0$, Rank $A = 3$, Nullity $A = 1$

(3) c_1, c_2, c_4 of A

(4) Nullsp $A = x_3 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$ basis for Nullsp $= \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$

(5) (a) F
(b) T
(c) F
(d) F

Version 4Q

(1) RREF $= \left[\begin{array}{cccc|cc} 1 & 0 & -1 & 0 & 0 & 2 \\ 0 & 1 & -1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$

(5) (a) F
(b) F
(c) T
(d) F